

Homework 5

Due: May. 11th

Turn in your homework in class

Rules

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism.
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit on time. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain 3 decimal places if rounding is required (unless otherwise specified).

1 Phasor

[12 points] Assuming the voltage and current of the two-terminal elements satisfy the passive sign convention (i.e., associated reference direction, where the current enters the positive terminal of the voltage), the instantaneous voltage and current are given as follows

(1) $v(t) = 12 \cos(200t + 20^\circ) \text{ V}$, $i(t) = 4 \sin(200t + 20^\circ) \text{ A}$;

(2) $v(t) = 6 \sin(300t + 40^\circ) \text{ V}$, $i(t) = 3 \sin(300t + 130^\circ) \text{ A}$;

(3) $v(t) = 10 \cos(100t + 45^\circ) \text{ V}$, $i(t) = 2 \sin(100t + 135^\circ) \text{ A}$;

(a) Transform the three voltage & current pairs into phasors.

(b) Try to determine whether the element is a resistor, inductor or capacitor, and determine its value ($R = ?$, $C = ?$, $L = ?$) for (1), (2), and (3), respectively.

2

[12 points] Obtain the equivalent impedance Z_{eq} of the circuit in Fig.2.

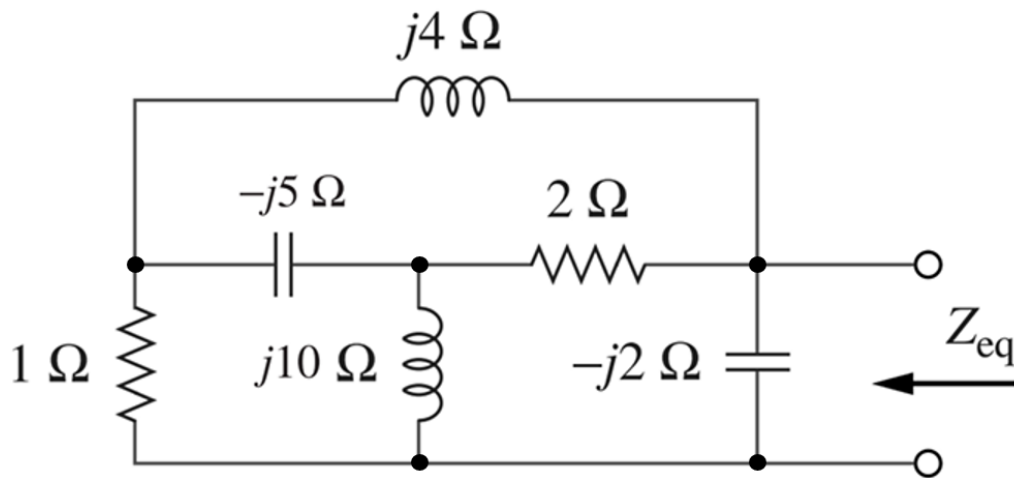
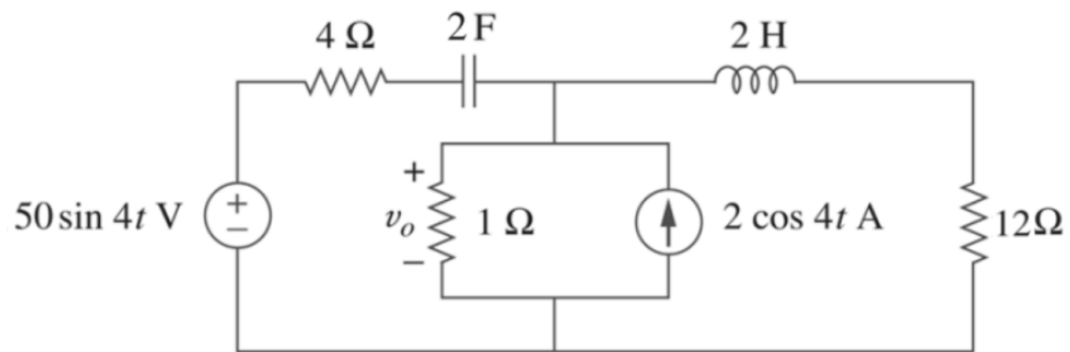


Fig.2

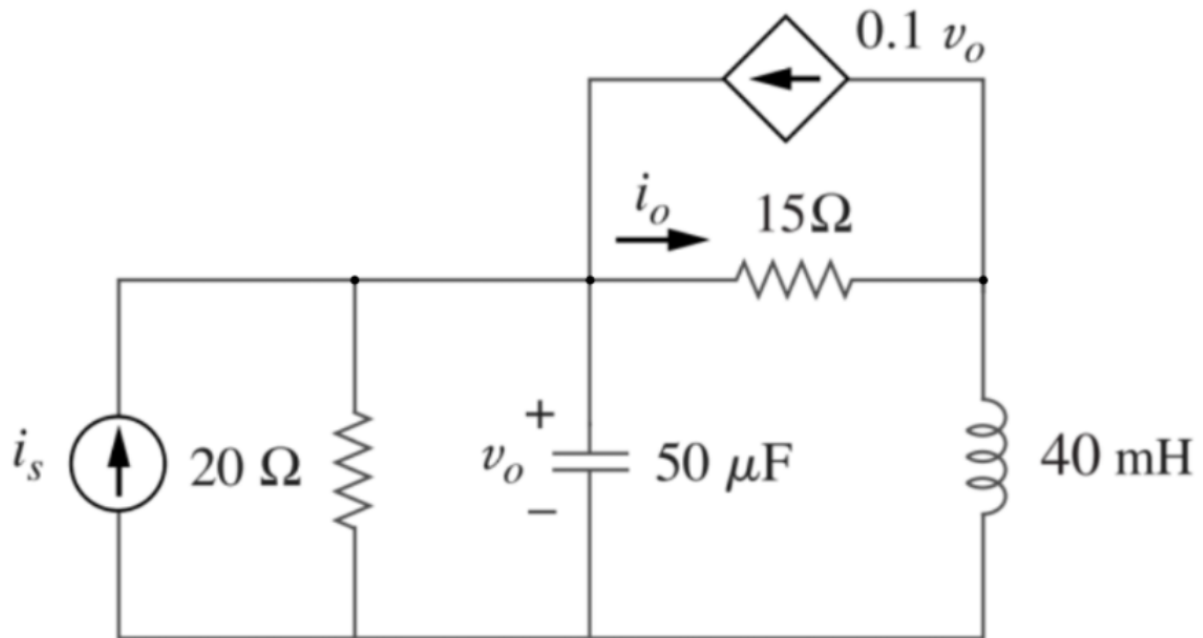
3

[13 points] Determine v_o in the circuit. The circuit is under sinusoidal steady state.



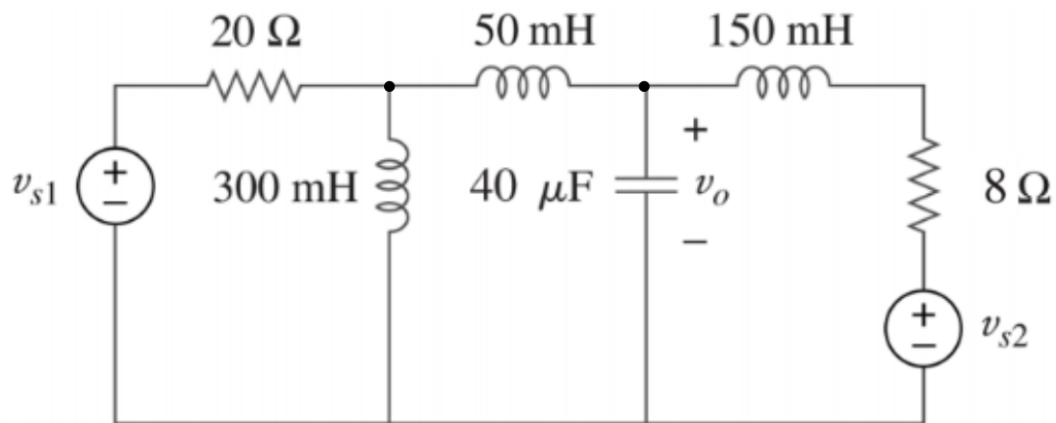
4 Nodal Analysis

[12 points] Use nodal analysis to find the current i_o in the circuit. Let $i_s = 6 \cos(200t + 15^\circ)$ A and the circuit is under sinusoidal steady state.



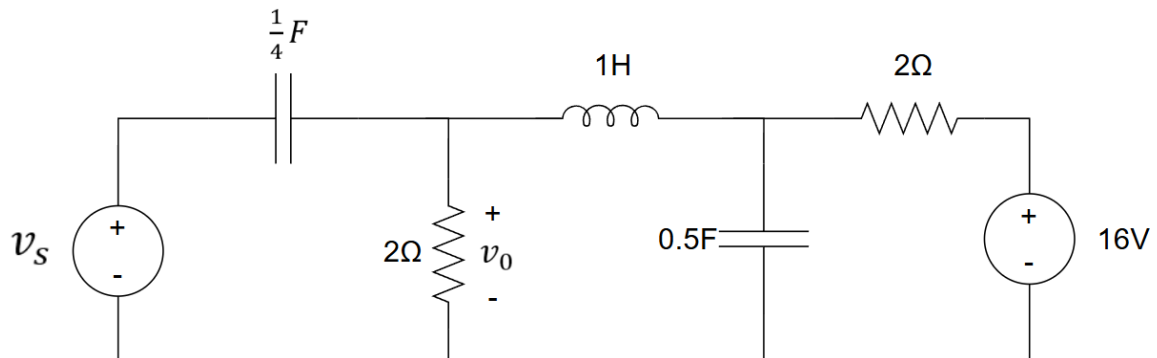
5 Mesh Analysis

[12 points] Use mesh analysis to find v_o in the circuit. Let $v_{s1} = 120 \cos(100t + 90^\circ)$ V, $v_{s2} = 80 \cos(100t)$ V and the circuit is under sinusoidal steady state.



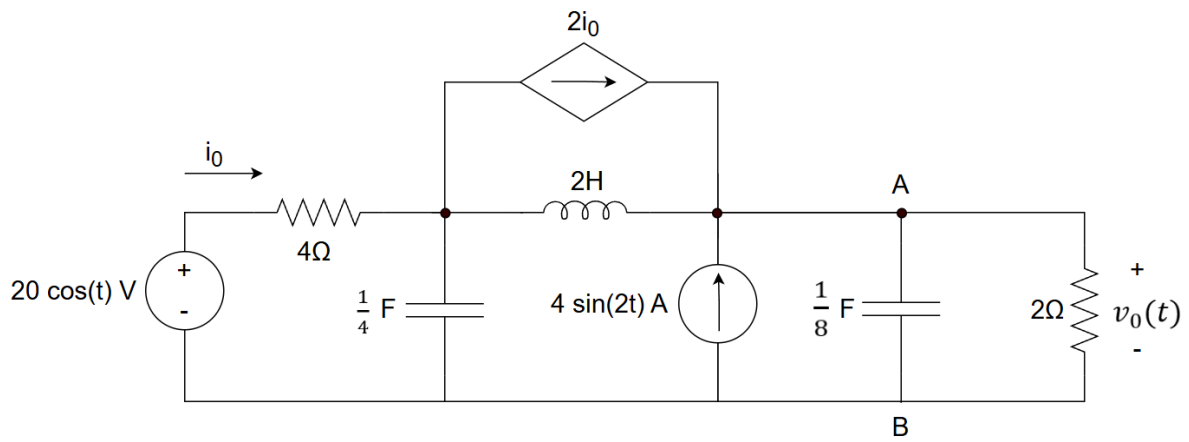
6 Superposition Theorem in AC circuits

[14 points] Solve $v_0(t)$ in the circuit below using the superposition principle. The circuit is under sinusoidal steady state and $v_s = 6\cos(2t) + 5\sin(4t)$ V.



7 Thevenin Equivalent AC circuits

[14 points] Use Thevenin's theorem at terminals A and B to find $v_0(t)$ in the circuit below. The circuit is under sinusoidal steady state.



8 Op Amp AC Circuits

[11 points] Compute $v_0(t)$ in the op amp circuit below if $v_s = 4\cos(1000t + 60^\circ)V$. The circuit is under sinusoidal steady state. Assume that the operational amplifiers are ideal and operate in the linear region.

